Predicting SpaceX Falcon 9 First-Stage Landing Success

IBM Applied Data Science Capstone

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Project repository: https://github.com/hongan151206-ai/Capstone-project

Executive Summary

- Goal: predict whether a Falcon 9 first stage lands successfully, because reuse is what lets SpaceX price a launch at ~$62M versus $165M+ for competitors.
- Methods: data from the SpaceX REST API and Wikipedia web scraping; wrangling with pandas (Class label); EDA with SQL and Seaborn/Matplotlib; Folium map and Plotly Dash dashboard; four classifiers (Logistic Regression, SVM, Decision Tree, KNN) tuned with GridSearchCV.
- Key results: 90 Falcon 9 launches, overall landing success rate 66.7%; success rose from 0.0% in 2010 to 84.2% in 2020; best model: SVM with 83.3% test accuracy.
- Conclusion: landing outcome is predictable from pre-launch features; flight experience, orbit and launch site drive success. Illustrative expected launch cost from predicted landing probability: ~$94.8M.

GitHub: https://github.com/hongan151206-ai/Capstone-project

Introduction

Project background

- Commercial space travel is growing and SpaceX leads the field thanks to reusable rockets.
- A Falcon 9 launch is advertised at ~$62M; other providers charge upwards of $165M. Most of the saving comes from reusing the first stage.

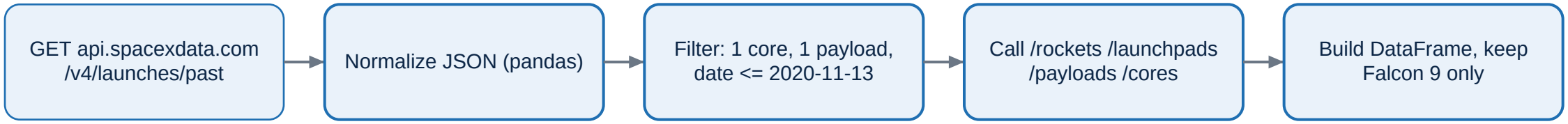
Problem statement

- We work as a data scientist for a company that wants to bid against SpaceX for a rocket launch.
- Task: determine the price of each launch by predicting whether the first stage will land successfully (and so be reused).
- Questions: what factors determine landing success? how did success change over time? which classifier predicts best?

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Data Collection - SpaceX API

Flowchart



- Key phrases: requests.get() -> .json() -> json_normalize() -> helper functions per endpoint
- Helpers: getBoosterVersion, getLaunchSite, getPayloadData, getCoreData
- Missing PayloadMass replaced with the column mean; Falcon 1 launches removed; FlightNumber re-indexed
- Result: dataset_part_1.csv with 90 Falcon 9 launches and 17 columns (Orbit, LaunchSite, Outcome, GridFins, Reused, Legs, ...)
- Code: 01_spacex_api_collection.py

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Data Collection - Web Scraping

Flowchart



- Key phrases: requests.get() -> BeautifulSoup() -> find_all('table') -> parse <th>/<td> cells
- Fields: flight no., launch site, payload, mass, orbit, customer, launch outcome, booster version, landing outcome, date, time
- Helpers clean date/time, booster version, landing status and payload mass strings
- Output: spacex_web_scraped.csv; used to cross-check the API data
- Code: 02_web_scraping.py

Data Wrangling Methodology

- Checked missing values and data types per column.
- Counted launches per site, per orbit and each landing outcome (True/False + Ocean, RTLS, ASDS; None = not attempted).
- Label: Class = 0 for bad outcomes (False Ocean, False RTLS, False ASDS, None ASDS, None None), otherwise 1.
- Feature table: one-hot encoding of Orbit, LaunchSite, LandingPad, Serial -> dataset_part_3.csv.
- Code: 03_data_wrangling.py, 05_eda_visualization.py

Results on the cleaned data

Launches: 90 | success rate (Class=1): 66.7%

Launches per site:

CCSFS SLC 40: 55

KSC LC 39A: 22

VAFB SLC 4E: 13

Top orbits: GTO (27), ISS (21), VLEO (14), PO (9), LEO (7)

Landing outcomes: True ASDS (41), None None (19), True RTLS (14), False ASDS (6), True Ocean (5)

Missing values: LandingPad (26)

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EDA with Data Visualization - Methodology

- Scatter plots (Seaborn catplot, hue = Class): Flight Number vs Payload Mass; Flight Number vs Launch Site; Payload Mass vs Launch Site.
- Scatter plots: Flight Number vs Orbit; Payload Mass vs Orbit - orbit-specific behaviour.
- Bar chart: success rate per orbit - which orbit types are easiest to land.
- Line chart: yearly average success rate - learning curve over time.
- Purpose: reveal how each feature relates to landing success, choose features for the models, and spot sparse categories.
- Code: 05_eda_visualization.py

EDA with SQL - Queries Performed

- Loaded Spacex.csv into a SQLite table SPACEXTBL and ran 10 queries:
  - Q1-Q2: unique launch sites; 5 records for sites starting with 'CCA'
  - Q3-Q4: total payload for NASA (CRS); average payload for booster F9 v1.1
  - Q5-Q6: first successful ground-pad landing; boosters with drone-ship success and 4000-6000 kg payload
  - Q7-Q8: successful vs failed missions; boosters carrying the maximum payload
  - Q9-Q10: failed drone-ship landings in 2015; ranking of landing outcomes 2010-06-04 to 2017-03-20
- Code: 04_eda_sql.py (results saved as sql_results/q1..q10.csv)

Interactive Visual Analytics - Folium and Plotly Dash

Folium map (06_folium_map.py)

- Circle + label for each launch site; marker clusters coloured green (success) / red (failure).
- MousePosition to pick coordinates; PolyLines with distance (haversine) to coastline, railway, highway and nearest city.

Plotly Dash app (07_dash_app.py)

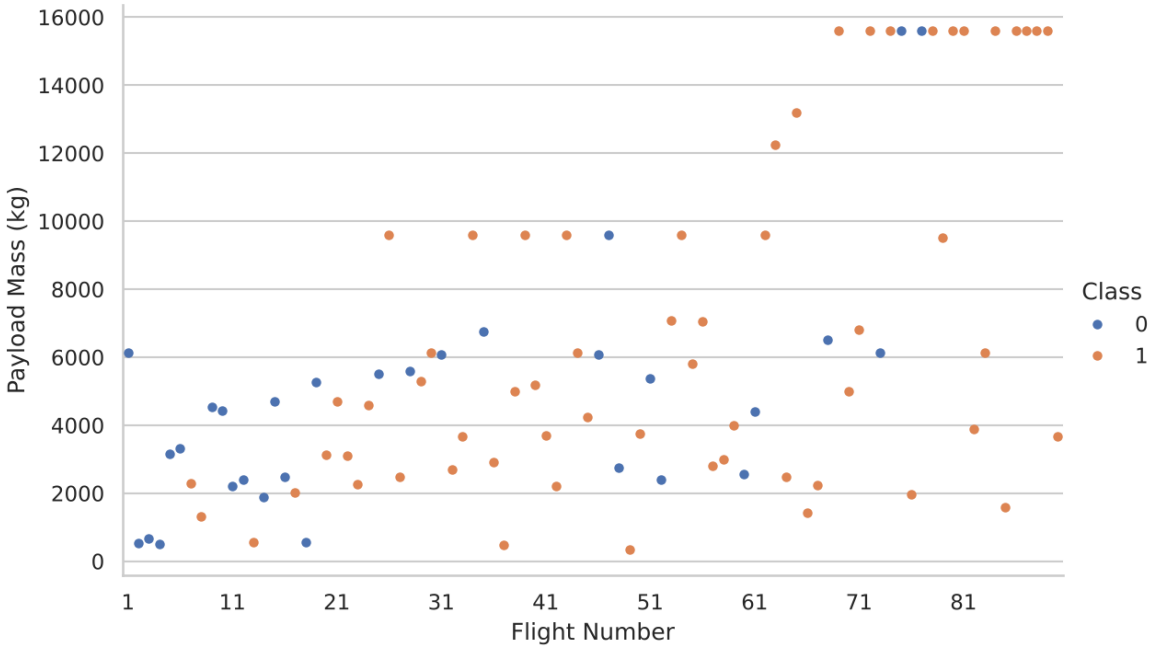
- Dropdown (all sites or one site) drives a pie chart of successful launches / success vs failure.
- Range slider filters payload mass; scatter plot of payload vs outcome coloured by booster version category.

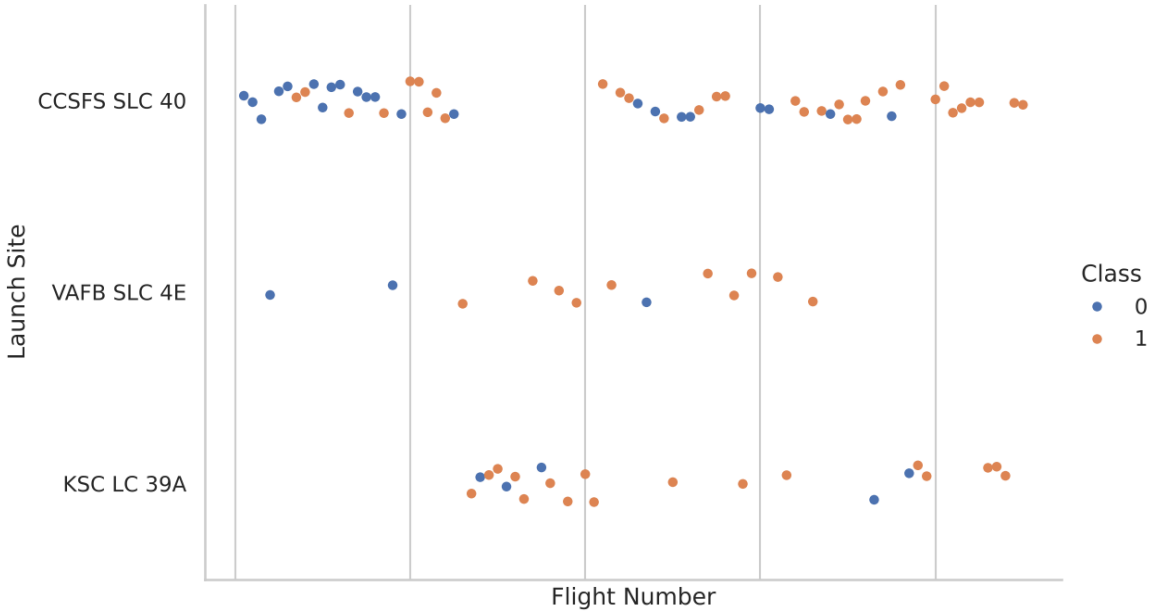
GitHub (Folium + Dash code): https://github.com/hongan151206-ai/Capstone-project

Predictive Analysis (Classification) - Methodology

- Target: Class (1 = first stage landed). Features: dataset_part_3 (one-hot encoded), standardised with StandardScaler.
- Split: 80% train / 20% test (random_state = 2) -> 72 train, 18 test launches.
- Models: Logistic Regression, Support Vector Machine, Decision Tree, K-Nearest Neighbours.
- Tuning: GridSearchCV with 10-fold cross-validation (C/penalty; kernel/C/gamma; criterion/depth/leaf; n_neighbors/algorithm/p).
- Evaluation: test accuracy, precision, recall, F1 and confusion matrix; best model chosen on test accuracy (ties -> CV accuracy).
- Code: 08_predictive_analysis.py

EDA Results - Scatter Plots (Flight Number, Payload, Site)

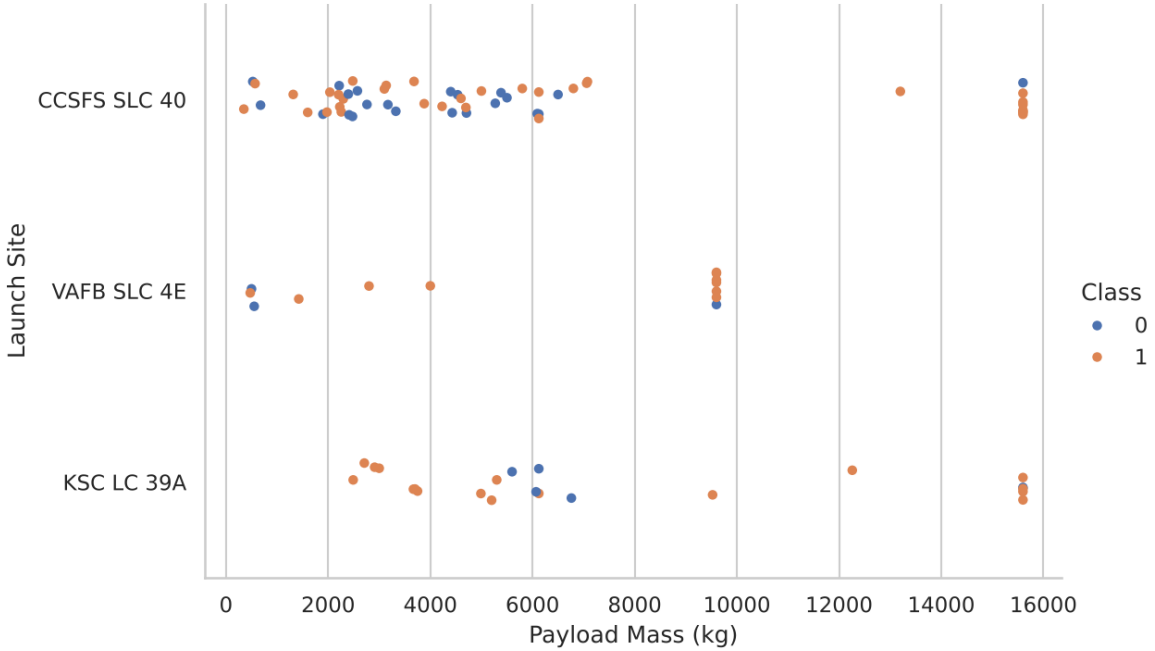


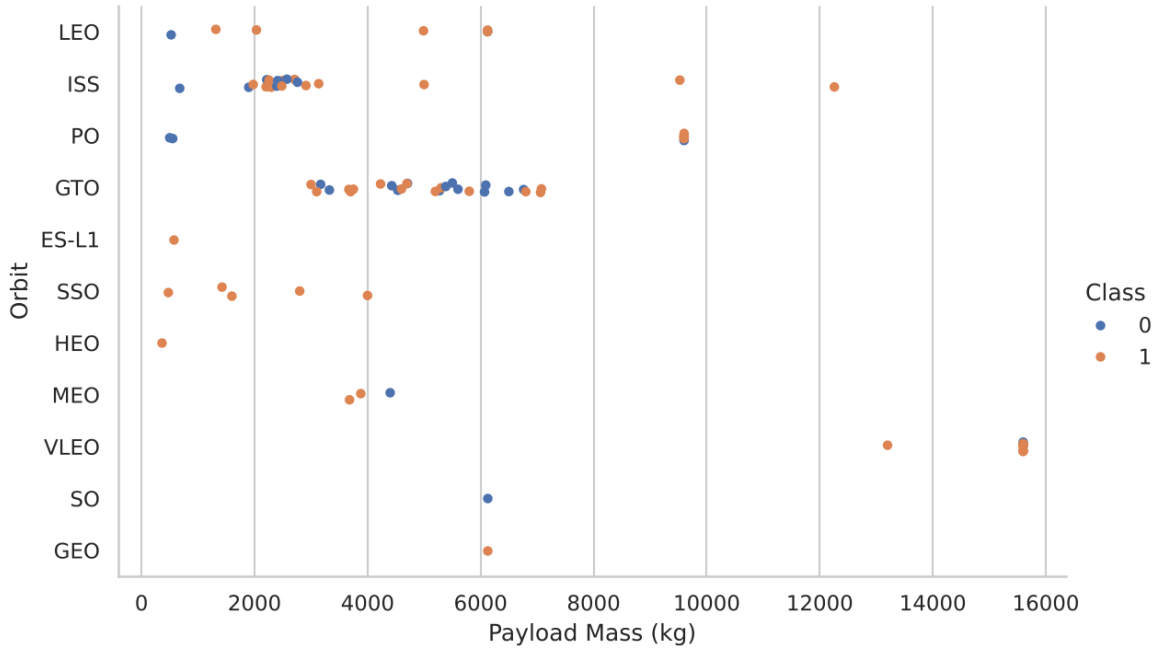


Success rate is 57.8% for flights 1-45 and 75.6% after that. Payloads under 5000 kg land 63.8% of the time vs 69.8% for heavier loads. Best site: KSC LC 39A.

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EDA Results - Payload by Site and Orbit

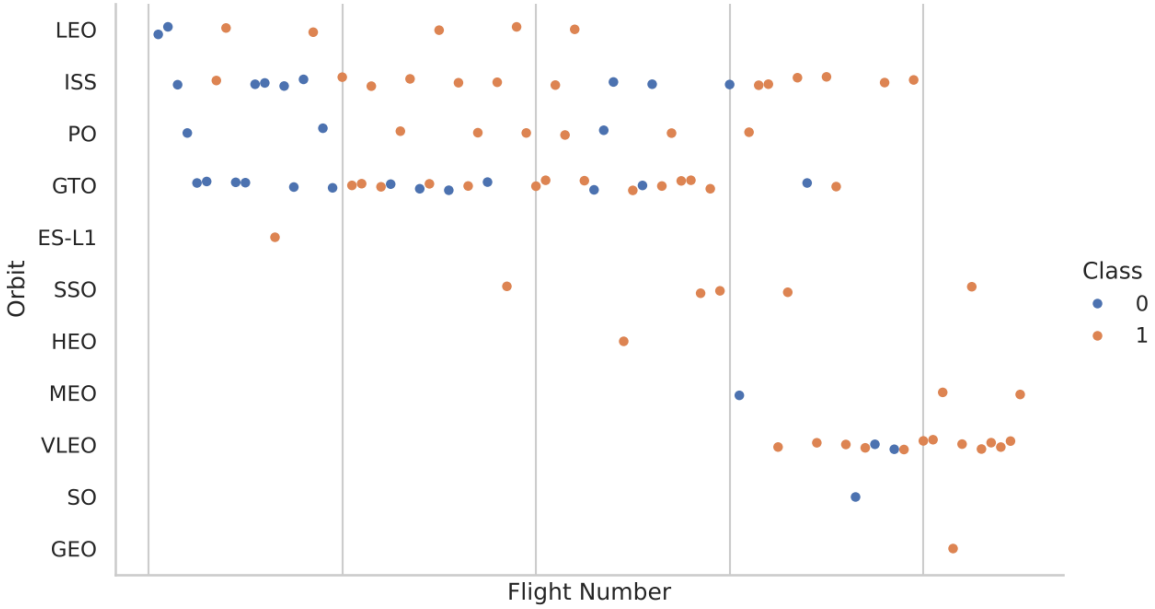


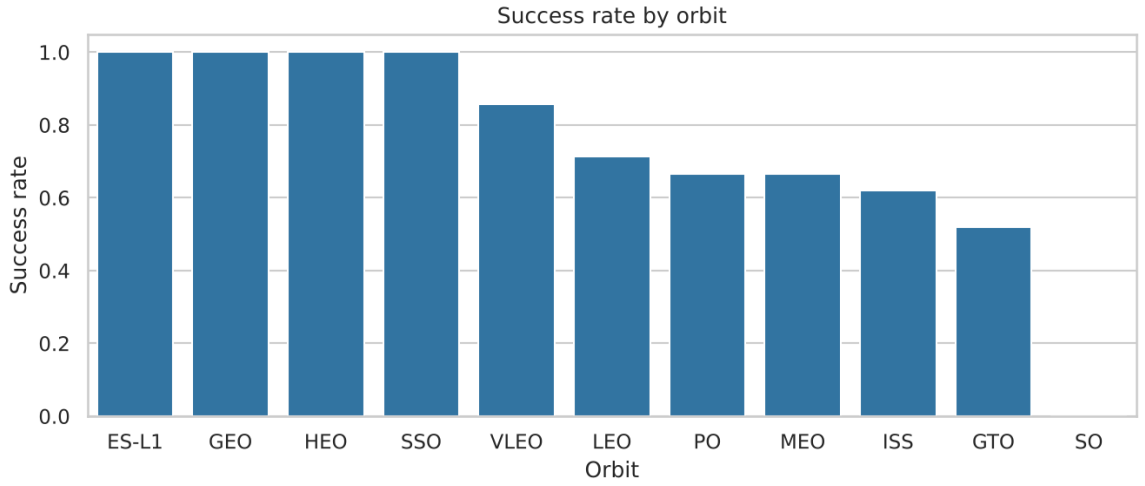


Payloads under 5000 kg: 63.8% success; 5000 kg and above: 69.8%. Best site by success rate: KSC LC 39A; weakest: CCSFS SLC 40.

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EDA Results - Orbit Analysis

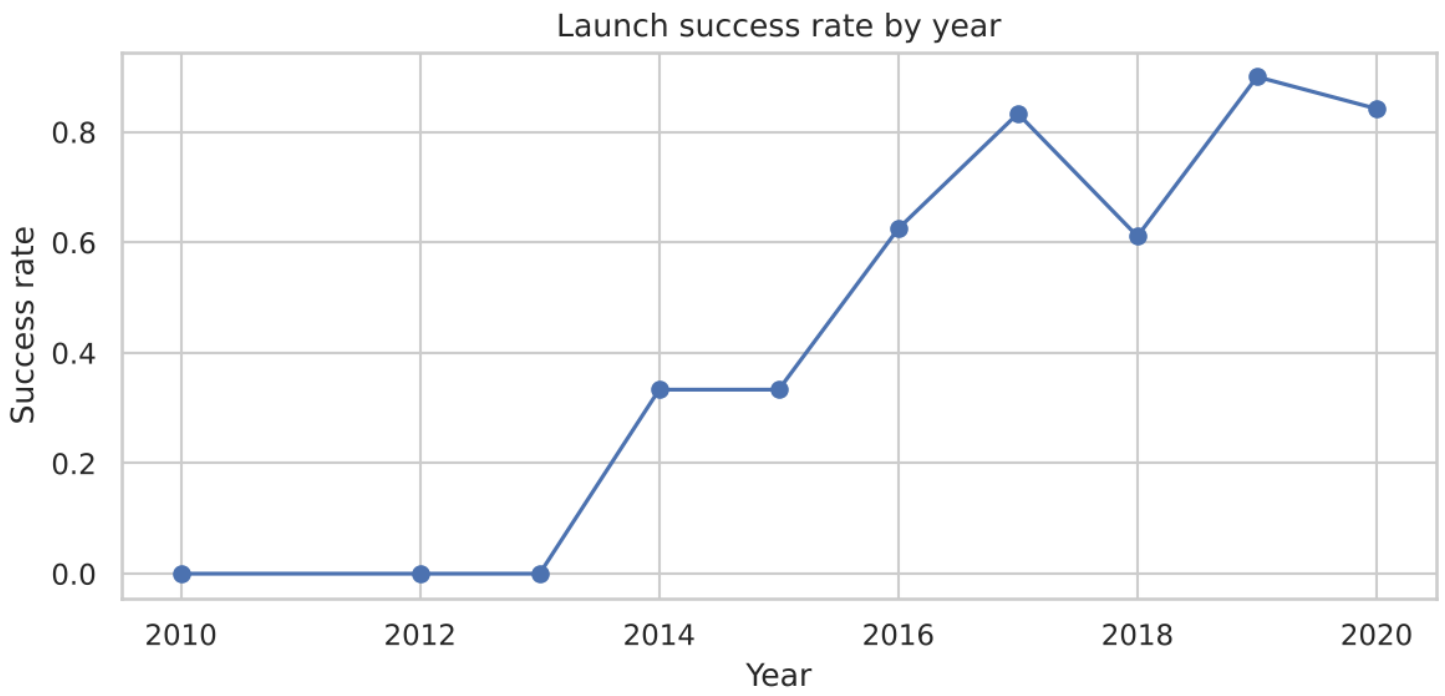




Orbits with 100% success have very few launches (ES-L1: 1, GEO: 1, HEO: 1, SSO: 5); among orbits with at least 8 launches the best is VLEO and the weakest is GTO.

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EDA Results - Yearly Trend



Success rate by year (launches)

2010: 0.0% (1)
2012: 0.0% (1)
2013: 0.0% (3)
2014: 33.3% (6)
2015: 33.3% (6)
2016: 62.5% (8)
2017: 83.3% (18)
2018: 61.1% (18)
2019: 90.0% (10)
2020: 84.2% (19)

Success rate moved from 0.0% (2010) to 84.2% (2020), peaking at 90.0% in 2019.

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EDA with SQL - Launch Sites and Payloads (Q1-Q4)

Q1 Unique launch sites

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Q2 Records for sites starting with 'CCA'

Date	Booster_Version	Launch_Site	Customer	Landing_Outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	NASA (COTS) NRO	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	NASA (CRS)	No attempt

Q3 Total payload mass, NASA (CRS)

total_kg
45596

Q4 Average payload mass, booster F9 v1.1

avg_kg
2,534.7

NASA (CRS) carried 45596 kg in total; boosters F9 v1.1 averaged 2,535 kg per launch.

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EDA with SQL - Landing Success and Mission Outcomes (Q5-Q8)

Q5 First successful ground-pad landing

MIN(Date)
2015-12-22

Q6 Drone-ship success, payload 4000-6000 kg

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Q7 Successful vs failed missions

Mission_Outcome	n
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Q8 Boosters carrying the maximum payload

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2

... +4 more rows (see sql_results/q8.csv)

First successful ground-pad landing: 2015-12-22. Boosters that landed on a drone ship with 4000-6000 kg payloads and the boosters that carried the maximum payload are listed above.

EDA with SQL - Rankings and Time Analysis (Q9-Q10)

Q9 2015 failed drone-ship landings

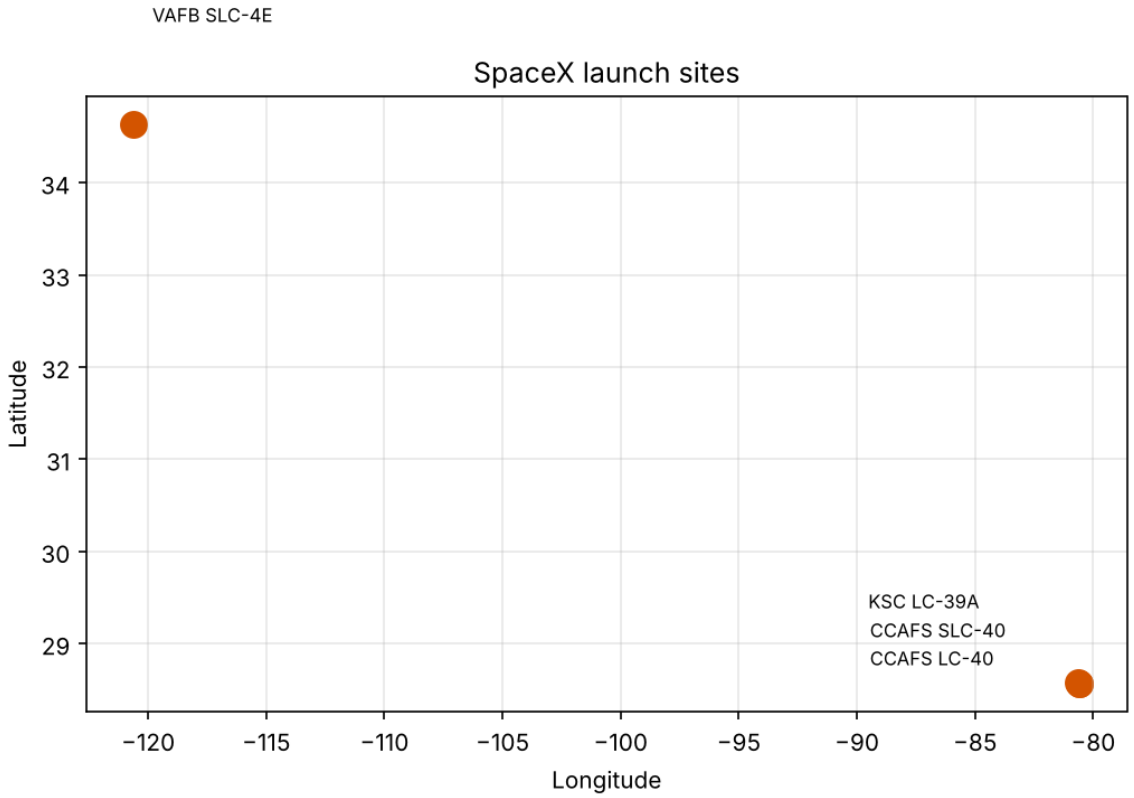
month	Booster_Version	Launch_Site
1	F9 v1.1 B1012	CCAFS LC-40
4	F9 v1.1 B1015	CCAFS LC-40

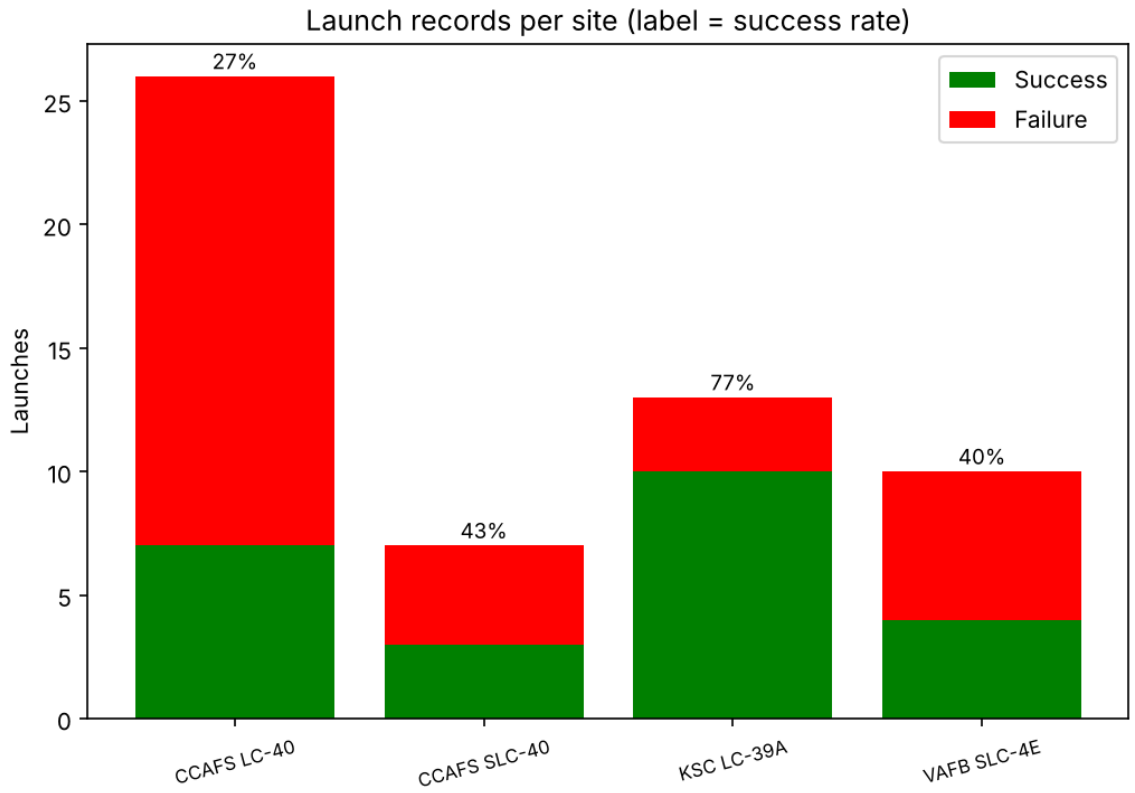
Q10 Landing outcomes ranked, 2010-06-04 to 2017-03-20

Landing_Outcome	n
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Most frequent outcome between 2010-06-04 and 2017-03-20: 'No attempt' (10 launches). Failed drone-ship landings in 2015 versus successful landings later show the landing technology maturing over time.

Folium Map - Launch Sites and Launch Records

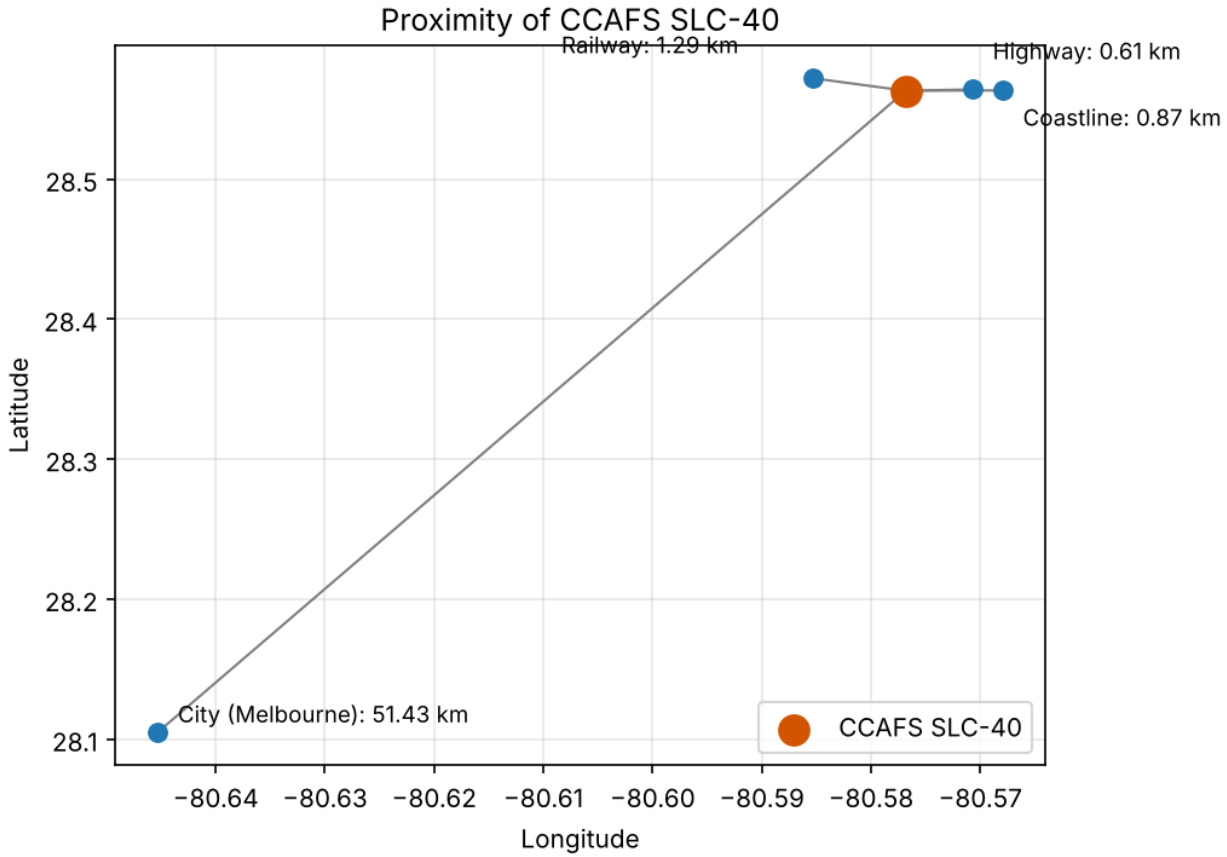




Launch sites: CCAFS LC-40 7/26 (26.9%); CCAFS SLC-40 3/7 (42.9%); KSC LC-39A 10/13 (76.9%); VAFB SLC-4E 4/10 (40.0%). Sites are on the US coasts (Florida, California); clusters are coloured green/red per launch.

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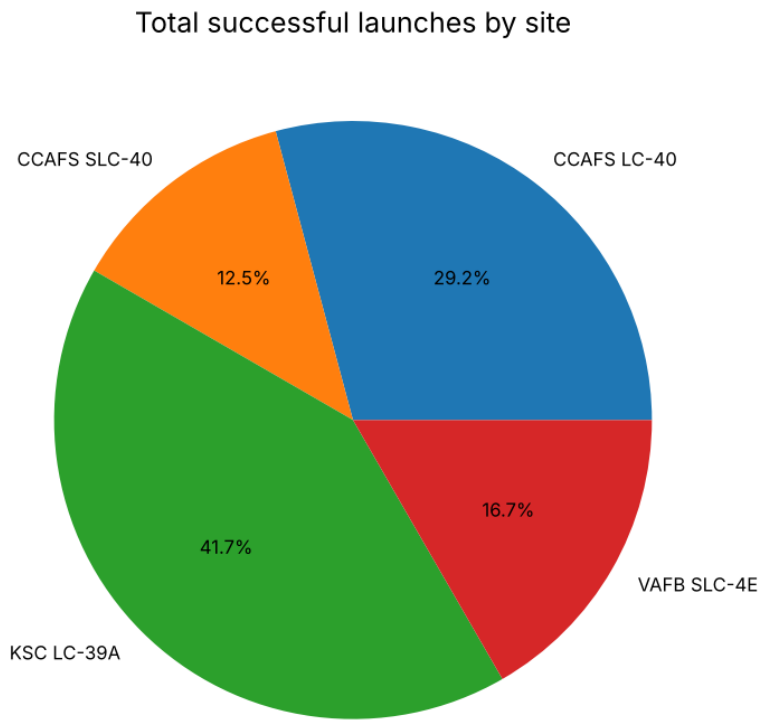
Folium Map - Proximity Analysis

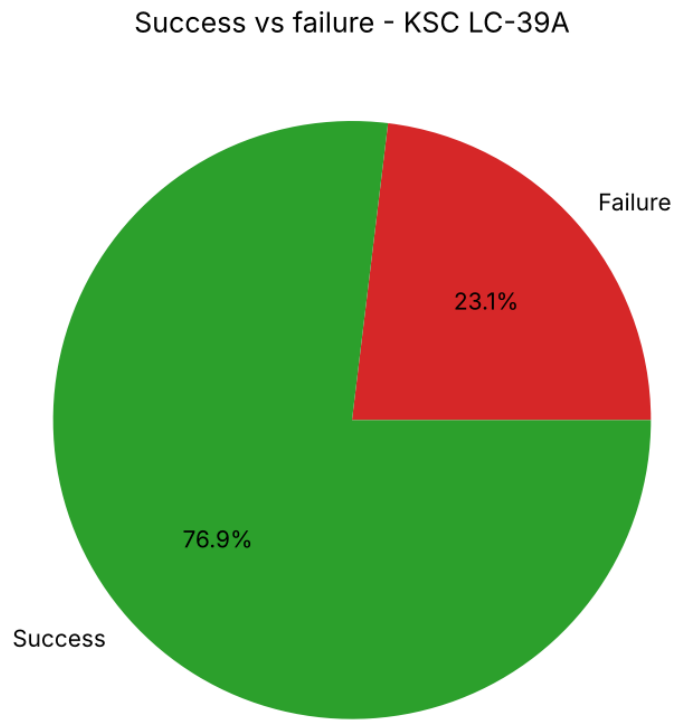


- Coastline: 0.87 km
- Railway: 1.29 km
- Highway: 0.61 km
- City (Melbourne): 51.43 km
- Close to coast, railway and highway: easy transport of rocket parts and safe launches over the ocean.
- Far from cities: limits risk to the population.

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Plotly Dash - Success Pie Charts

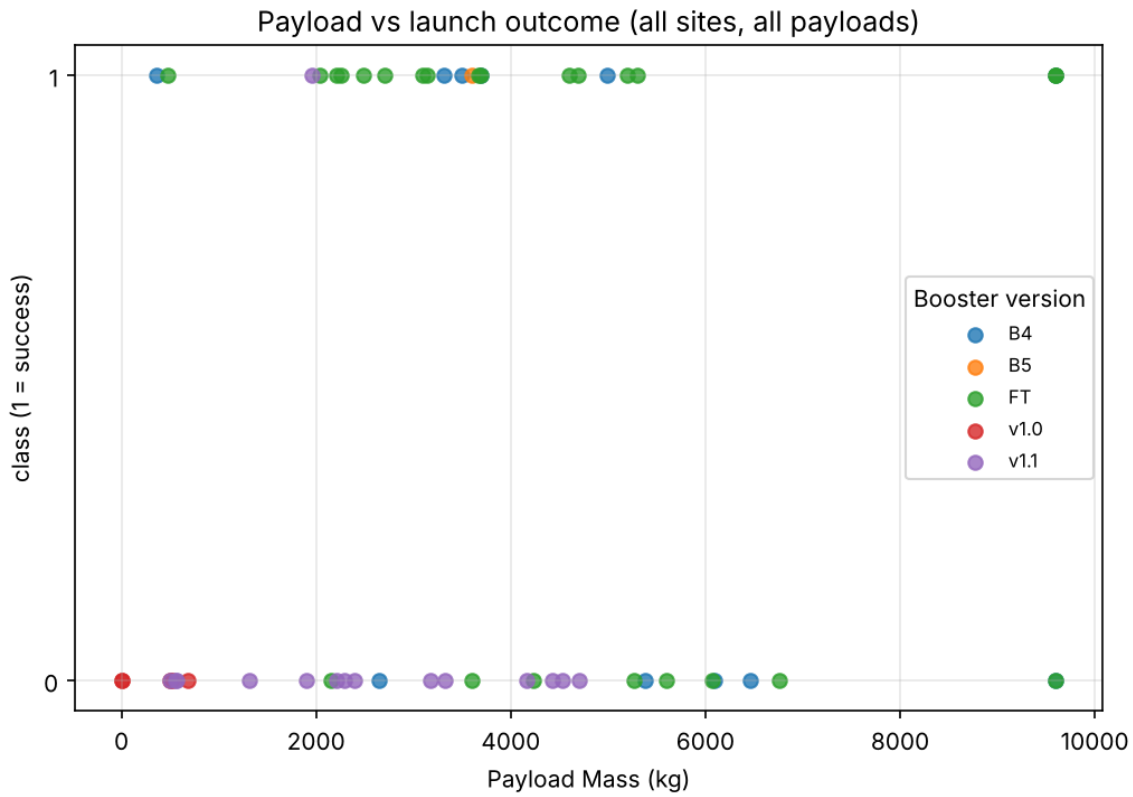


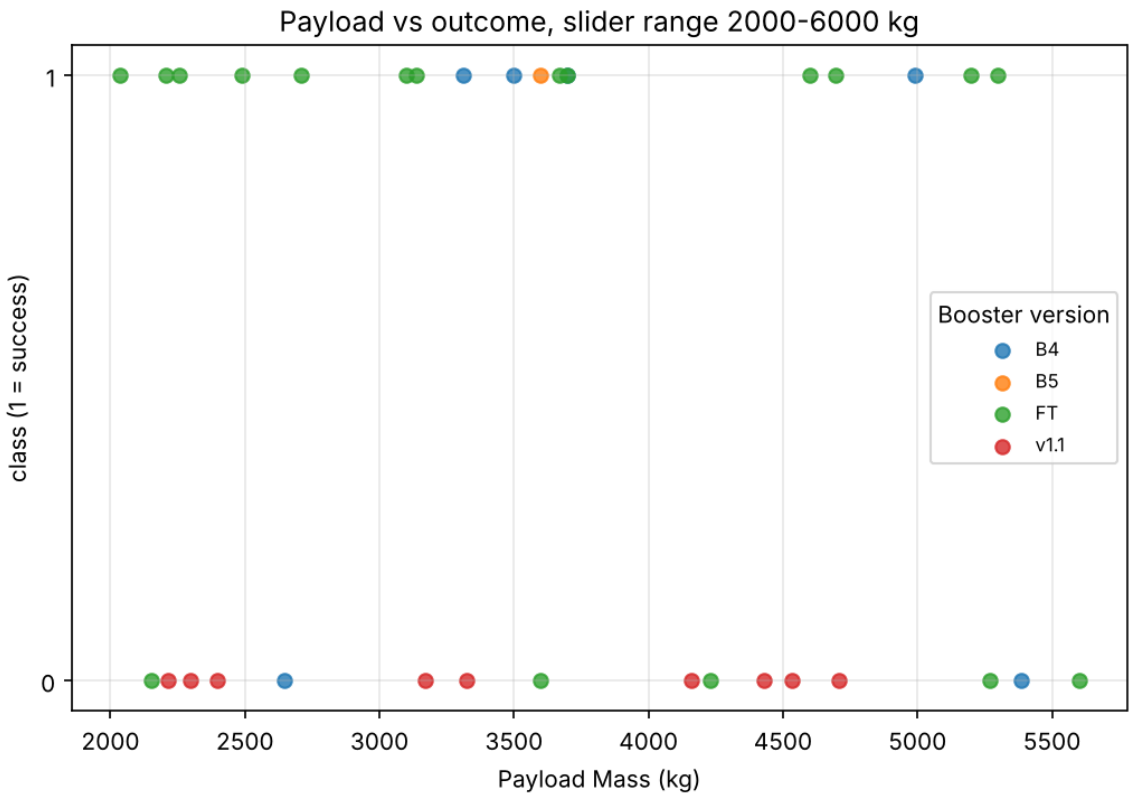


KSC LC-39A accounts for the largest share of successful launches; KSC LC-39A has the best success ratio (right pie).

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Plotly Dash - Payload vs Launch Outcome

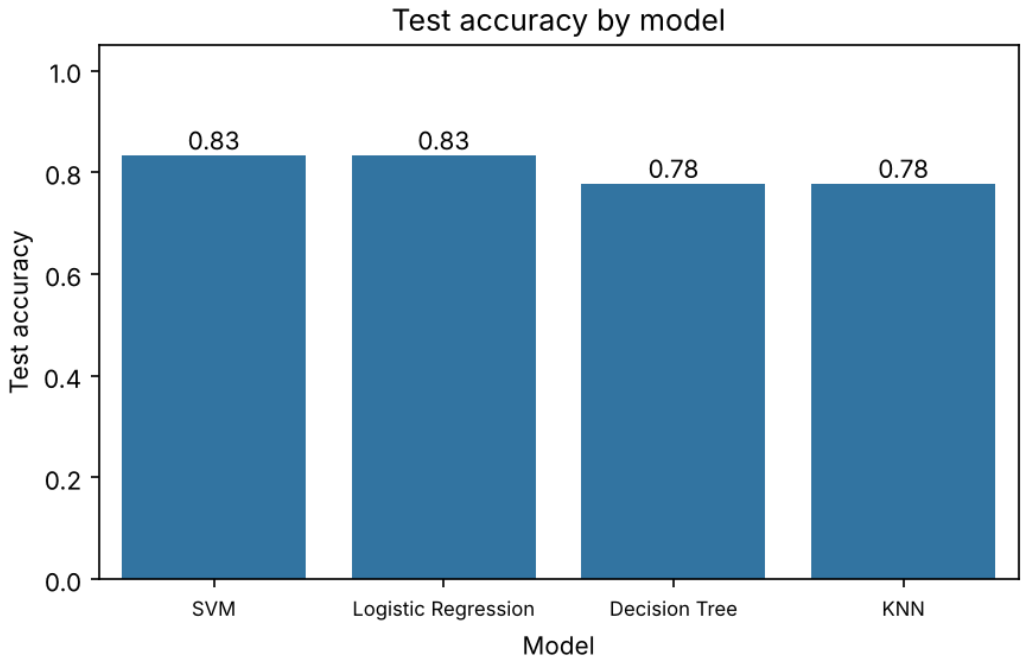




Success rate is 45.2% for payloads under 5000 kg vs 35.7% above; by booster version: B4 54.5%, B5 100.0%, FT 66.7%, v1.0 0.0%, v1.1 6.7%.

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Predictive Analysis Results - Model Comparison

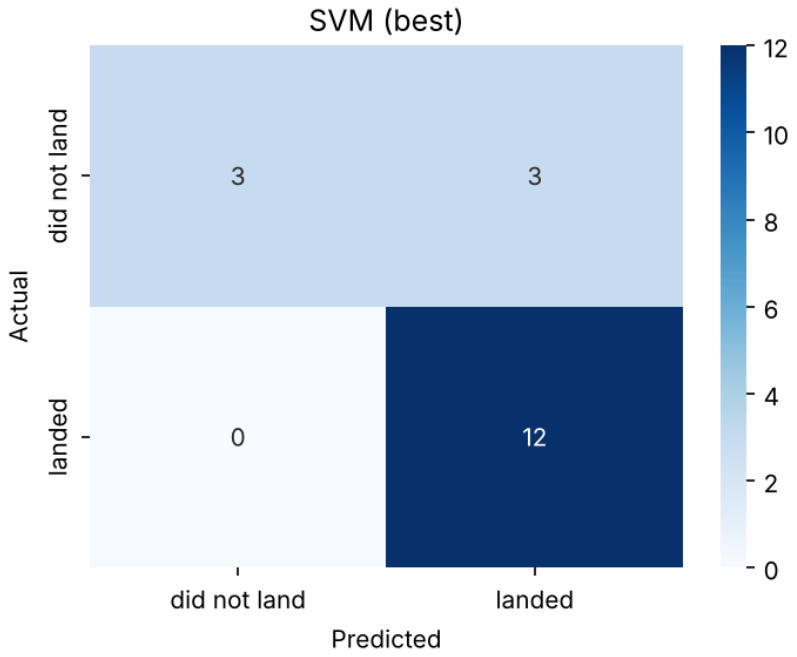


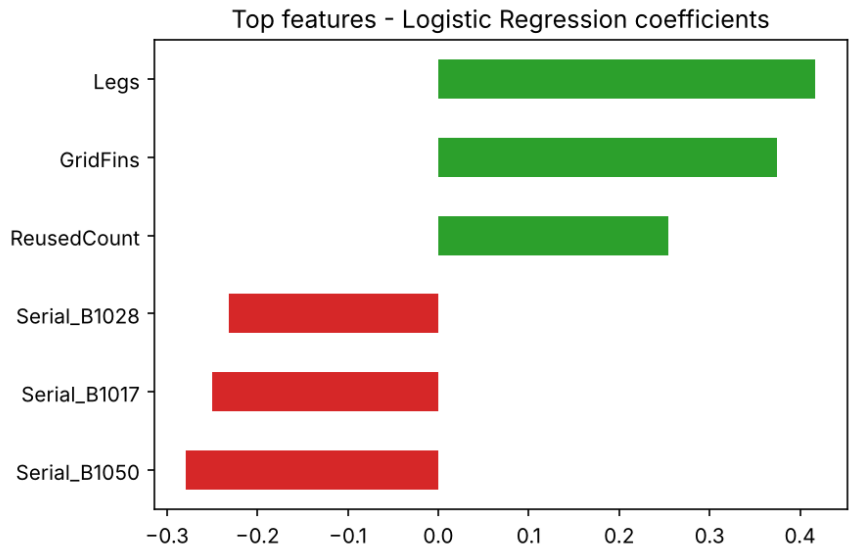
Model	CV acc	Test acc	Precision	Recall	F1
SVM	0.848	0.833	0.800	1.000	0.889
Logistic Regression	0.821	0.833	0.800	1.000	0.889
Decision Tree	0.836	0.778	0.900	0.750	0.818
KNN	0.834	0.778	0.750	1.000	0.857

- Best model: SVM - test accuracy 83.3%
- Best parameters: {'C': 1.0, 'gamma': 0.03162277660168379, 'kernel': 'sigmoid'}

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Predictive Analysis Results - Confusion Matrix and Best Model





- SVM: TP 12, TN 3, FP 3, FN 0 on 18 test launches.
- FP = predicted a landing that did not happen; FN = missed a successful landing.
- Strongest features (logistic regression coefficients): Legs, GridFins, Serial_B1050, ReusedCount, Serial_B1017, Serial_B1028.

Test set is small (18 launches), so differences between models are within noise; compare the models on the F1 score and the confusion matrix as well as accuracy.

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Conclusion and Innovative Insights

- Landing success is predictable from pre-launch data: SVM reaches 83.3% test accuracy.
- Success changed from 0.0% to 84.2% across years; later flights succeed 75.6% of the time vs 57.8% for early ones.
- Orbit and site matter: best site KSC LC 39A; payloads below 5000 kg land 63.8% vs 69.8%.
- Innovative insight - cost model: converting predicted landing probability into expected cost (assumption: $62.0M if landed, $165.0M if not) gives about $94.8M per launch, a basis for a competing bid.
- Next steps: add weather, booster age and landing-pad data; try ensembles (Random Forest, XGBoost); validate with launches after 2020.

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